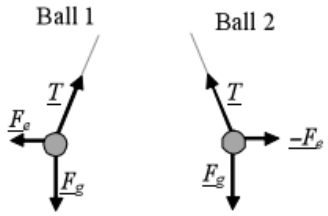
Week 1: Electrostatics, Gauss' Law and Potential Difference SOLUTIONS

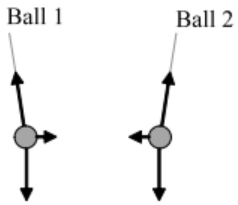
Discussion Questions

1a



1b Angle reduces but balls make same angle with vertical as each other

1c

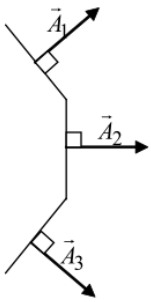


They attract because the charged ball induces separation of charges on the uncharged ball.
(If they were to touch they would share the available charge, so each would get q/2- and then they would repel)

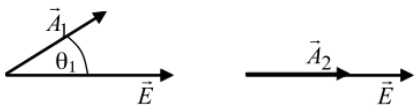
2a. forces are radially outwards from the position of the rod
2b. force on A about twice the force on ball at B (E field is not inversely proportional to R^2; closer to inversely proportional to R)
2c. would be halved **2d** field would increase ($\propto Q$ on rod); field would not change; No, does not affect the field as it is the charge on the *rod* that establishes the field..

3

a.



b. The flux through the top-third surface is *less than* the flux through the middle-third surface, because the electric field component normal to the top-third surface is smaller than the electric field component normal to the middle-third surface, even though the areas of the top-third and middle-third surfaces are the same by construction (we folded the paper into three equal parts). Mathematically we have $\vec{E} \bullet \vec{A}_1 = EA_1 \cos\theta_1$ and $\vec{E} \bullet \vec{A}_2 = EA_2 \cos\theta_2 = EA_1$ since $|\vec{A}_1| = |\vec{A}_2|$ (paper folded into three equal parts) and $\cos\theta_2 = 1$ ($\theta_2 = 0$). Needless to say, $EA_1 \cos\theta_1$ is less than EA_1.



c. $\Phi_{net} = \vec{E} \bullet \vec{A}_1 + \vec{E} \bullet \vec{A}_2 + \vec{E} \bullet \vec{A}_3 = EA_1(\cos\theta_1 + 1 + \cos\theta_1) = EA_1(1 + 2\cos\theta_1)$

4a no charge inside surface hence total flux = 0

4b $\Phi_L + \Phi_C = 0$ hence $\Phi_C = -\Phi_L$

4c **(i)** Φ_L does not change; distance of charge from dark shaded surface is the same

(ii) Same again (flux into closed surface is still equal to flux out since there is no charge inside the surface)

4d **(i)** Φ_L still does not change, distances and area all the same as in part (c)

(ii) this must increase since total flux through whole surface is now >0

(iii) $\Phi_C = q/\varepsilon_0 - \Phi_L$ (follows from total flux = q/ε_0)

e i. Greater than. The electric field increases in magnitude, because two charges are of the same sign are now generating it (principle of superposition). Hence, Φ_L increases, even though the area of the left-hand portion of the surface is unchanged.

ii. Equal to. By Gauss's law, the total flux is proportional to the total charge enclosed by the surface, which is unchanged.

5 (d) because electric field in the conductor must be zero (static case) and hence no work is required to move a charge anywhere in the conductor.

Problem Solving Questions

6. Forces are attractive. Use Coulomb's Law ($F = kq_1q_2/r^2$) = 2.81 N

The magnitude of the mutual force of attraction at $r = 0.120$ m is

$$F = k\frac{|q_1||q_2|}{r^2} = (8.99 \times 10^9)\frac{(3.00 \times 10^{-6})(1.50 \times 10^{-6})}{0.120^2} = 2.81 \text{ N}.$$

7.

By symmetry we see the contributions from the $+q$ charges cancel each other, and we simply use Eq. 23-3 to compute magnitude of the field due to the $+2q$ charge (this field points at 45°, which is clear from the figure in the textbook).

$$\left|\vec{E}_{\text{net}}\right| = \frac{1}{4\pi\varepsilon_0}\frac{2q}{r^2}$$

where $r = a/\sqrt{2}$. Thus, we obtain $\left|\vec{E}_{\text{net}}\right| = q/\pi\varepsilon_0 a^2$.

8.

(a) The linear charge density is the charge per unit length of rod. Since the charge is uniformly distributed on the rod, $\lambda = -q/L$.

(lambda's units are C/m)

(b) We position the x axis along the rod with the origin at the left end of the rod, as shown in the diagram. Let dx be an infinitesimal length of rod at x. The charge in this segment is $dq = \lambda\, dx$. The charge dq may be considered to be a point charge. The electric field it produces at point P has only an x component and this component is given by

$$dE_x = \frac{1}{4\pi\varepsilon_0}\frac{\lambda\, dx}{(L+a-x)^2}.$$

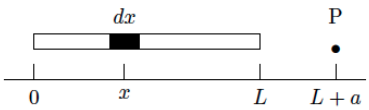
The total electric field produced at P by the whole rod is the integral

$$\begin{aligned} E_x &= \frac{\lambda}{4\pi\varepsilon_0}\int_0^L \frac{dx}{(L+a-x)^2} \\ &= \frac{\lambda}{4\pi\varepsilon_0}\left.\frac{1}{L+a-x}\right|_0^L \\ &= \frac{\lambda}{4\pi\varepsilon_0}\left(\frac{1}{a} - \frac{1}{L+a}\right) \\ &= \frac{\lambda}{4\pi\varepsilon_0}\frac{L}{a(L+a)}. \end{aligned}$$

When $-q/L$ is substituted for λ the result is

$$E_x = -\frac{1}{4\pi\varepsilon_0}\frac{q}{a(L+a)} .$$

The negative sign indicates that the field is toward the rod.



(c) If a is much larger than L, the quantity $L + a$ in the denominator can be approximated by a and the expression for the electric field becomes

$$E_x = -\frac{q}{4\pi\varepsilon_0 a^2} .$$

9 Total flux through the cube surfaces = q/ε_0 hence through one surface = (1/6) q/ε_0

10

We imagine a cylindrical Gaussian surface A of radius r and unit length concentric with the metal tube. Then by symmetry

$$\oint_A \vec{E}\cdot d\vec{A} = 2\pi r E = \frac{q_{\text{enclosed}}}{\varepsilon_0} .$$

(a) For $r > R$, $q_{\text{enclosed}} = \lambda$, so $E(r) = \lambda/2\pi r\varepsilon_0$.

(b) For $r < R$, we note that $q_{\text{enclosed}} = 0$. If there is an E field in this region, it must be cylindrically symmetric. Any cylindrical Gaussian surface with an axis common to the other cylinders (length l, radius r) must have $2\pi rlE = q_{\text{enclosed}}/\varepsilon_0$; hence $E = 0$ everywhere.

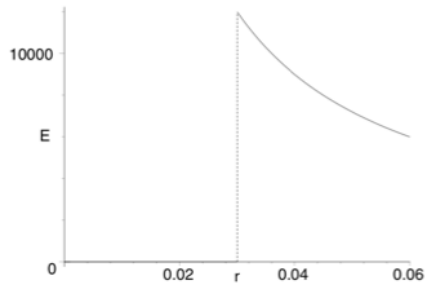
We imagine a cylindrical Gaussian surface A of radius r and unit length concentric with the metal tube. Then by symmetry

$$\oint_A \vec{E}\cdot d\vec{A} = 2\pi r E = \frac{q_{\text{enclosed}}}{\varepsilon_0} .$$

(a) For $r > R$, $q_{\text{enclosed}} = \lambda$, so $E(r) = \lambda/2\pi r\varepsilon_0$.

(b) The plot of E vs r is shown below. Here, the maximum value is

$$E_{\max} = \frac{\lambda}{2\pi r\varepsilon_0} = \frac{(2.0\times10^{-8}\,\text{C/m})}{2\pi(0.030\,\text{m})\,(8.85\times10^{-12}\,\text{C}^2/\text{N}\cdot\text{m}^2)} = 1.2\times10^4\ \text{N/C} .$$



11

We imagine a spherical Gaussian surface of radius r centered at the point charge $+q$. From symmetry consideration E is the same throughout the surface, so

$$\oint \vec{E}\cdot d\vec{A} = 4\pi r^2 E = \frac{q_{\text{encl}}}{\varepsilon_0} ,$$

hence

$$E(r) = \frac{q_{\text{encl}}}{4\pi\varepsilon_0 r^2} ,$$

where q_{encl} is the net charge enclosed by the Gaussian surface.

(a) Now $a < r < b$, where $E = 0$. Thus $q_{\text{encl}} = 0$, so the charge on the inner surface of the shell is $q_i = -q$.

(b) The shell as a whole is electrically neutral, so the outer shell must carry a charge of $q_o = +q$.

(c) For $r < a$ $\quad q_{\text{encl}} = +q$, so

$$E\Big|_{r<a} = \frac{q}{4\pi\varepsilon_0 r^2} .$$

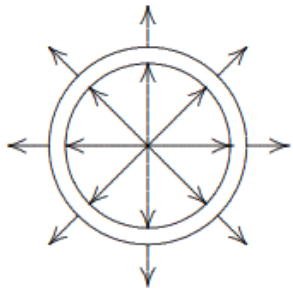
12

(d) For $b > r > a$ $E = 0$, since this region is inside the metallic part of the shell.

(e) For $r > b$ $q_{\text{encl}} = +q$, so

$$E\Big|_{r<a} = \frac{q}{4\pi\varepsilon_0 r^2}\ .$$

The field lines are sketched to the right.



(f) The net charge of the central point charge-inner surface combination is zero. Thus the electric field it produces is also zero.

(g) The outer shell has a spherically symmetric charge distribution with a net charge $+q$. Thus the field it produces for $r > b$ is $E = q/(4\pi\varepsilon_0 r^2)$.

The contributions to the potential from the $+q$ at a distance d and the $-q$ at a distance d cancel (no vectors here!). Hence just sum the contributions from the remaining two charges i.e.

$$V_{total} = \frac{kq}{d} - \frac{kq}{2d} = \frac{kq}{2d} \quad \text{(this is 0.5 mV)}$$

13a

Distance to +3 nC charge must be half the distance to the –6 nC charge. Hence at (-2, 0) and at (-18, 0)

13b No. Any such points must be to the left of the +3 nC charge (say at $-x$, 0), and obey the equation:

$$\frac{3}{(x-6)^2} = \frac{6}{(x+6)^2}$$

the solution to this is x = 35 cm.

13c

Yes. There are lots of points that are twice the distance from the -6 nC charge compared to the +3 nC charge.

13d

Yes; the condition is that the distance to the 3 nC charge must be half the distance to the 6 nC charge. For example, there is a point at (-6, +4√3).

13d lots of them! (eg the 0 V equipotential line must go through (-4, 0) and (-18, 0); shape can be inferred from field lines (perpendicular to these)

Past Exam Questions

14. (2005)

ai attractive, 3.4 mN **aii** attractive, 3.4 mN **aiii** (-5.96**i**, +1.94**j**) mN (use Coulomb/components)
b Transfer of electrons from the wool to the comb; some polarization of small particles; polarization of the neutral particles leads to unequal repulsive and attractive forces, with the attractive force the stronger (needs a diagram); if the force was uniform all it would do would be to rotate the (polarized) particles, whilst in a non-uniform field there is also a net attraction.

15. (2006)

ai electrons removed **aii** 3.1×10^{10} electrons
b -11 nC (Coulomb's Law); **ci** to the left (vector addition); **cii** to the right (larger charge on B and always closer); **ciii** to the right (close to charge A) and then to the left (large charge on B) **civ** 1.82 m to left of A (balance fields vectorially)

16 (a) short solutions should give enough information.
16 (b) Create Gaussian surface (eg cylinder) bisecting plane of conductive sheet, with surface "caps" of area A equidistant from plane. Relate enclosed charge for surface to charge density by $q_{enc} = \sigma A$. Now solve for field using Gauss' Law: $\int E.dA = \frac{q_{enc}}{\varepsilon_0} \rightarrow$ $E \int dA = \frac{q_{enc}}{\varepsilon_0} \rightarrow E(A_{side1} + A_{side2}) = \frac{\sigma A}{\varepsilon_0} \rightarrow E = \frac{\sigma A}{2A\varepsilon_0} = \sigma/2\varepsilon_0$

17:

i Could say no electric fields in conductors. Or no net flux through conductors. Could show no enclosed charge, thus no electric field, using Gauss' Law. (for a<r<b this needs to show equilibrium reached, changes moved to lowest potential energy, so net charge for Gaussian "inside outer shell is zero).

ii this can be shown from a spherical Gaussian surface *inside* the outer conductor; it cannot have a net flux through it (since E = 0 everywhere), hence there cannot be any <u>net</u> charge inside it.

iii total charge must add up to -7Q, so the surface must have the remainder (-5Q)

iv standard theory here

v, vi and vii the wire establishes zero potential difference between all parts of the (now connected) conductor, hence there can be no field between the two shells, hence no charge on the inner conductor, hence the net charge will move to the outside of the outer sphere